

Lab 4.2.7 Report: Getting Familiar with the Linux Shell

Student Name: omar mohammad hamed

Part 1: Shell Basics

Step 2: Display Manual Pages

Question 1: Name a few sections that are included in a man page.

Answer: The sections included in a man page are:

- NAME: Command name and brief description
- SYNOPSIS: Command syntax and available options
- DESCRIPTION: Detailed explanation of command functionality
- OPTIONS: List of command options and their descriptions
- EXAMPLES: Practical usage examples
- SEE ALSO: Related commands and resources
- AUTHOR: Information about the command's author

Question 2: What is the function of the cp command?

Answer: The cp command is used to copy files and directories from one location to another in the Linux filesystem. It creates a duplicate of the source file at the destination while keeping the original file intact.

Question 3: What command would you use to find out more information about the pwd command? What is the function of the pwd command?

Answer:

- Command: man pwd
 - Function: The pwd (Print Working Directory) command displays the complete pathname of the current working directory, showing the user's current location in the filesystem hierarchy.
-

Step 3: Create and Change Directories

Question 1: What is the current directory?

Answer: /home/analyst

Question 2: Which folder are you in now?

Answer: cyops_folder3 (full path: /home/analyst/cyops_folder3)

Challenge Question: Type the command `cd ~` and describe what happens. Why did this happen?

Answer: When executing `cd ~`, the terminal returns to the home directory (/home/analyst). This happens because the tilde (~) symbol is a special shortcut in Linux that always represents the current user's home directory, providing a quick way to navigate back home from any location in the filesystem.

Question 3: What happens when using `cd .`?

Answer: Nothing visible happens because the dot (.) represents the current directory itself. The command essentially changes to the same directory, resulting in no actual movement in the filesystem.

Question 4: What happens when using `cd ..`?

Answer: The command changes to the parent directory (one level up in the directory hierarchy). From /home/analyst/cyops_folder3, it returns to /home/analyst.

Question 5: What would be the current directory if you issued the `cd ..` command at [analyst@secOps ~]\$?

Answer: /home (parent directory of analyst)

Question 6: What would be the current directory if you issued the `cd ..` command at [analyst@secOps home]\$?

Answer: / (root directory, parent of home)

Question 7: What would be the current directory if you issued the `cd ..` command at [analyst@secOps /]\$?

Answer: / (remains at root directory, as there is no parent directory above root)

Step 4: Redirect Outputs

Question 1: Is that expected? Explain.

Answer: Yes, this is expected behavior. When using the redirect operator (>), the command output is sent to the specified file instead of being displayed on the terminal screen. Since the output was redirected to some_text_file.txt, no text appears in the terminal window.

Question 2: What happened to the text file? Explain.

Answer: The original content of the file was completely overwritten with the new message. The single redirect operator (>) replaces all existing content in the target file with the new output. This is destructive operation that does not preserve previous data.

Step 5: Redirect and Append to a Text File

Question: What happened to the text file? Explain.

Answer: The new line was added to the end of the existing file content without deleting previous data. The double redirect operator (>>) appends data to the file rather than overwriting it, preserving all existing content and adding new content at the end.

Step 6: Work with Hidden Files in Linux

Question 1: How many files are displayed?

Answer: Approximately 8-10 items (files and directories) are displayed when using ls -l without the -a option.

Question 2: How many more files are displayed than before? Explain.

Answer: Approximately 15-20 additional files are displayed. These are hidden files whose names begin with a dot (.). Linux does not display dot-files by default unless the -a option is used with the ls command. Hidden files typically contain configuration data and user preferences.

Question 3: Is it possible to hide entire directories? Are there any directories in the output?

Answer: Yes, directories can be hidden by adding a dot before their names. Several hidden directories appear in the ls -la output, including .cache, .config, .local, and others. These directories are treated exactly like hidden files.

Question 4: Give three examples of hidden files shown in the output.

Answer:

1. .bashrc - Bash shell configuration file

2. `.bash_history` - Command history file
 3. `.profile` - User profile configuration file
-

Part 2: Copying, Deleting, and Moving Files

Step 1: Copying Files

Question: What are the source and destination files? (use full paths)

Answer:

- Source file: `/home/analyst/some_text_file.txt`
 - Destination file: `/home/analyst/cyops_folder2/some_text_file.txt`
-

Step 3: Moving Files and Directories

Question: What command did you use to accomplish the task?

Answer: Command used: `mv cyops_folder2/some_text_file.txt .`

The dot (`.`) at the end represents the current directory (`/home/analyst`), providing a convenient shorthand for specifying the destination. This command moves the file from `cyops_folder2` back to the home directory.

Reflection

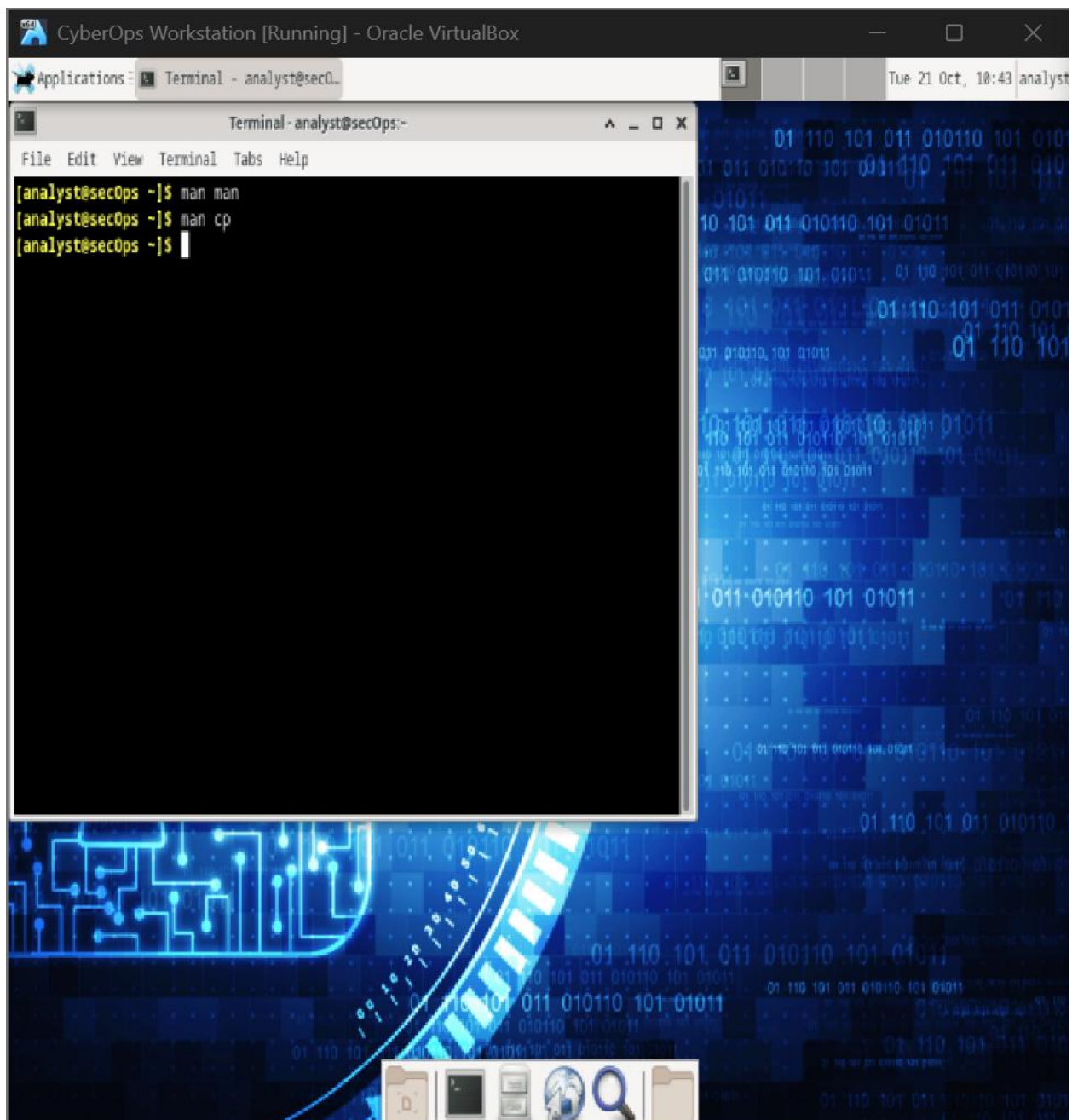
What are the advantages of using the Linux command line?

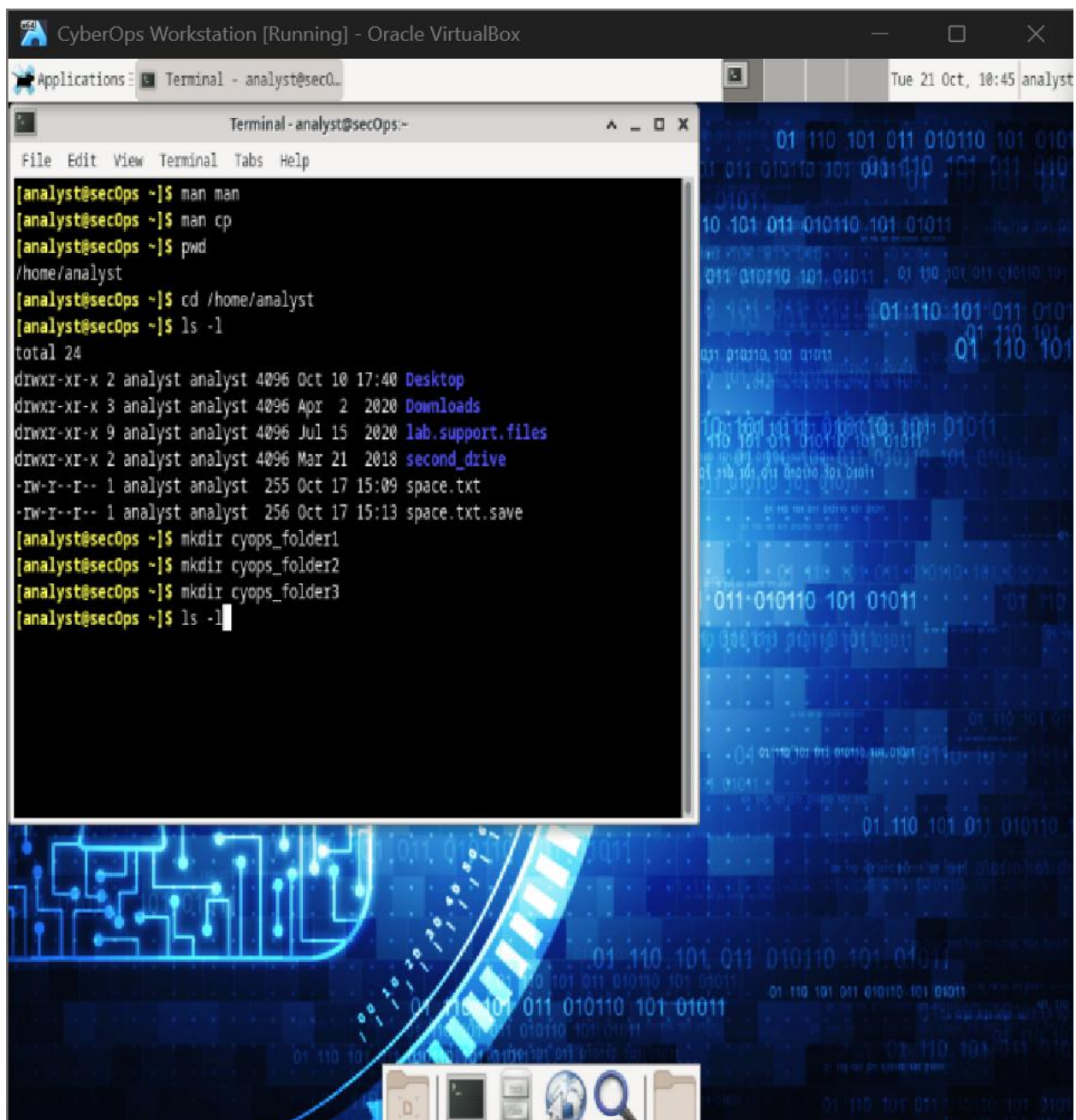
The Linux command line offers numerous significant advantages:

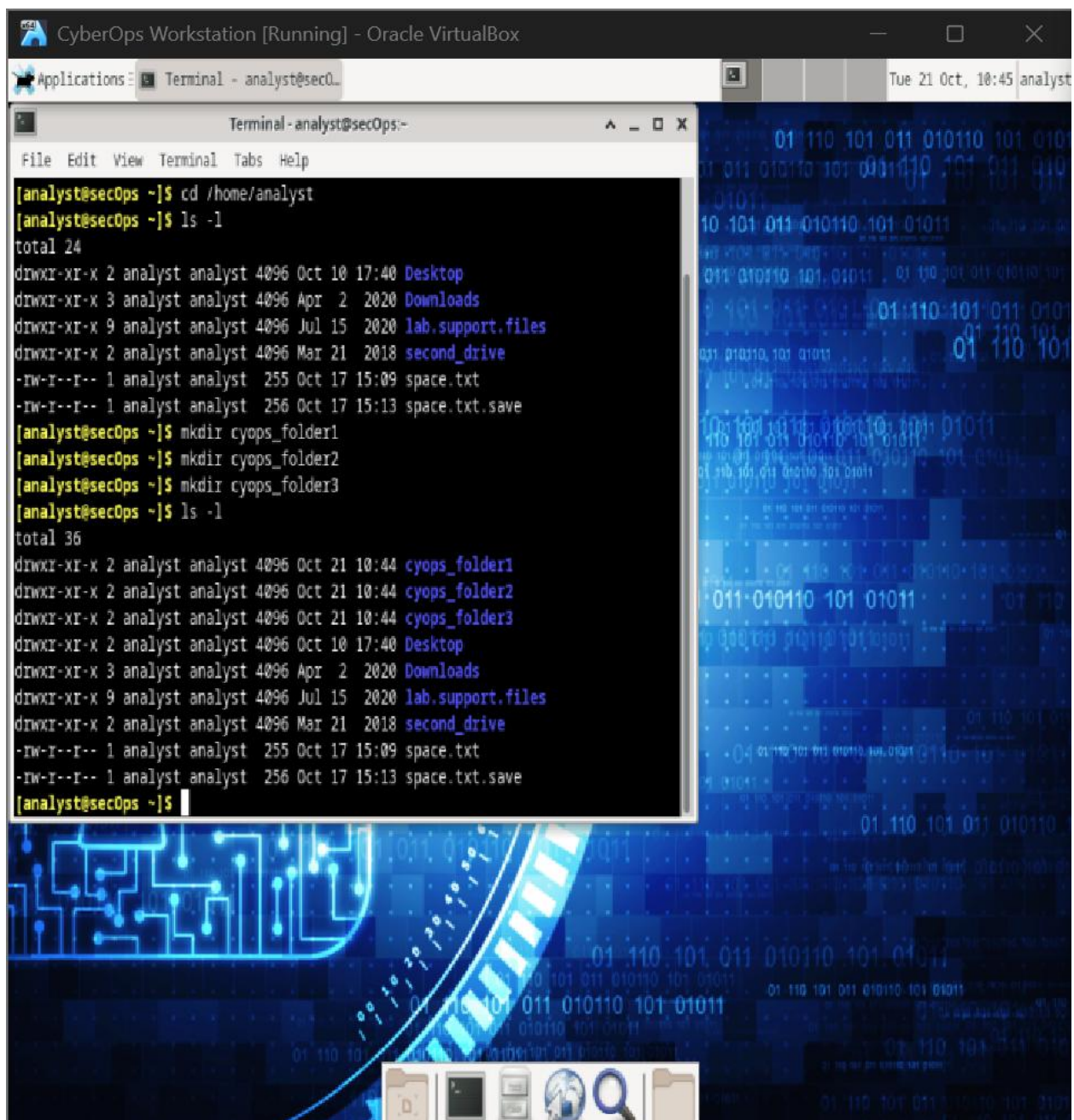
1. **Speed and Efficiency:** Commands execute much faster than equivalent GUI operations, especially for repetitive tasks or operations on multiple files.
2. **Automation Capability:** Shell scripts can automate complex sequences of commands, saving time and reducing human error in routine operations.
3. **Remote Management:** The command line enables efficient remote system administration through SSH, allowing management of servers without physical access.

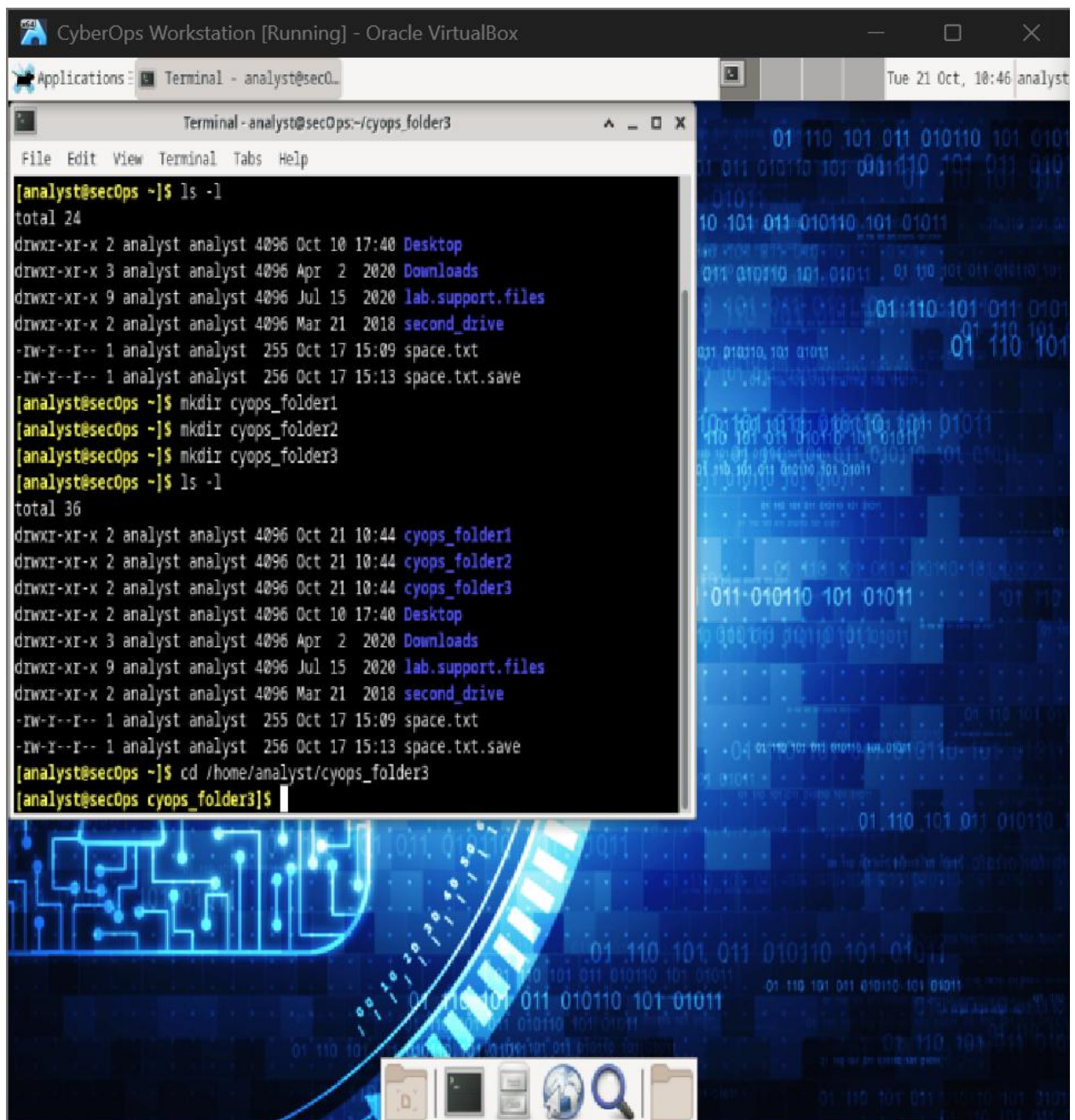
4. **Superior Control:** Provides more precise control over system operations, with access to advanced options not available in graphical interfaces.
 5. **Resource Efficiency:** Consumes minimal system resources compared to graphical interfaces, crucial for server environments and older hardware.
 6. **Batch Processing:** Can perform operations on hundreds or thousands of files simultaneously using wildcards and loops.
 7. **Professional Necessity:** Essential skill for system administrators, cybersecurity professionals, and DevOps engineers.
-

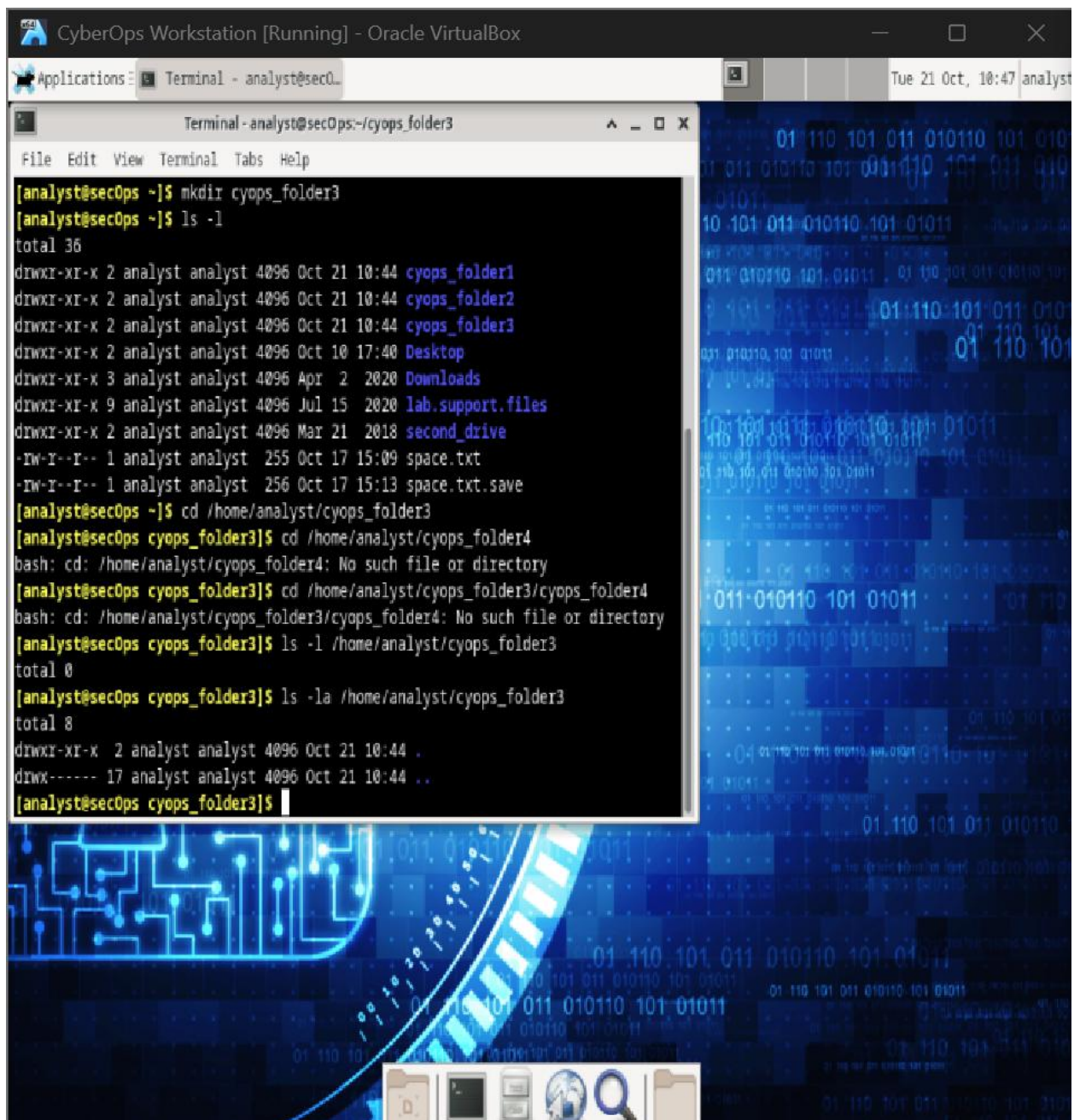
Screenshots:

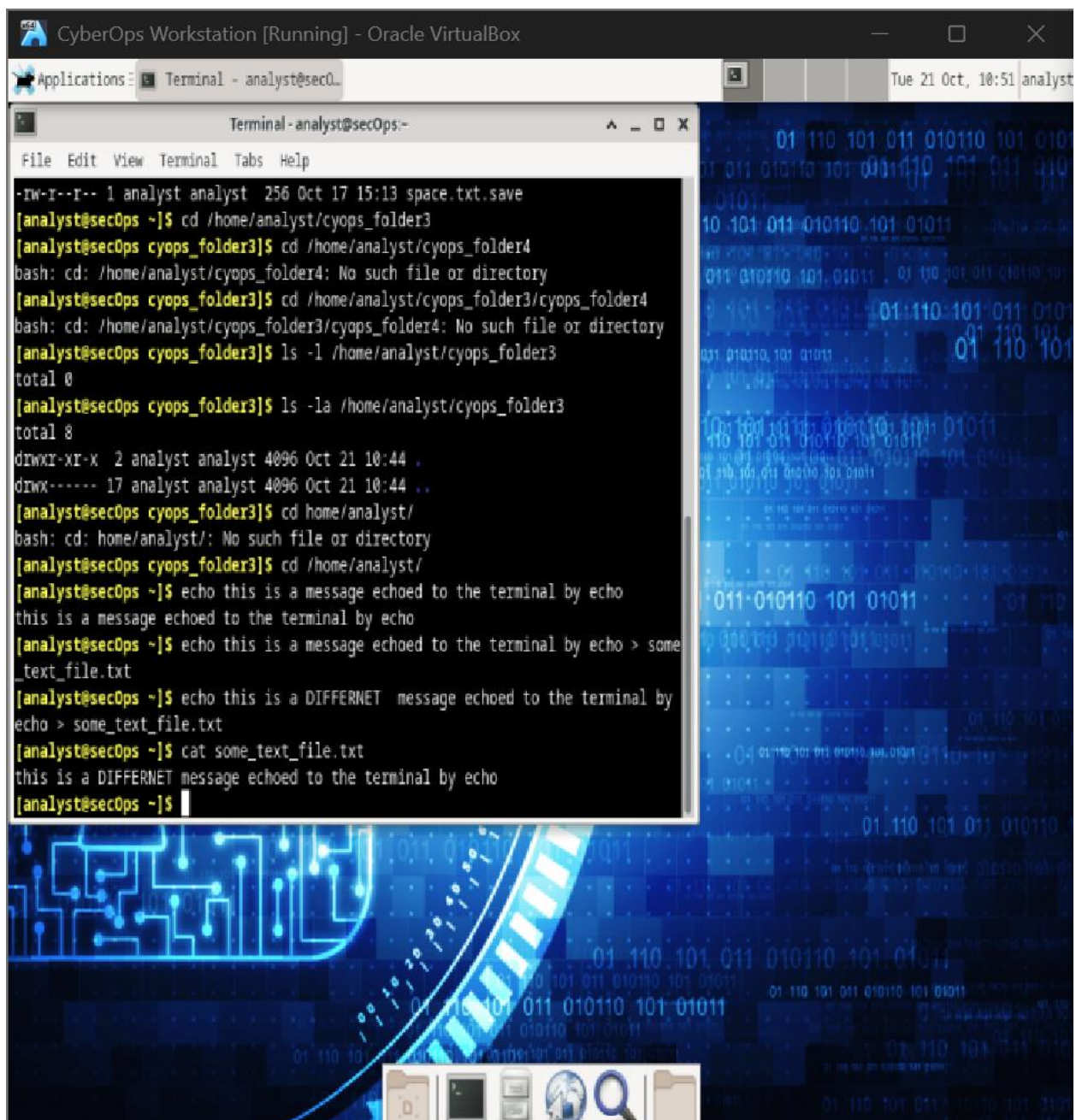


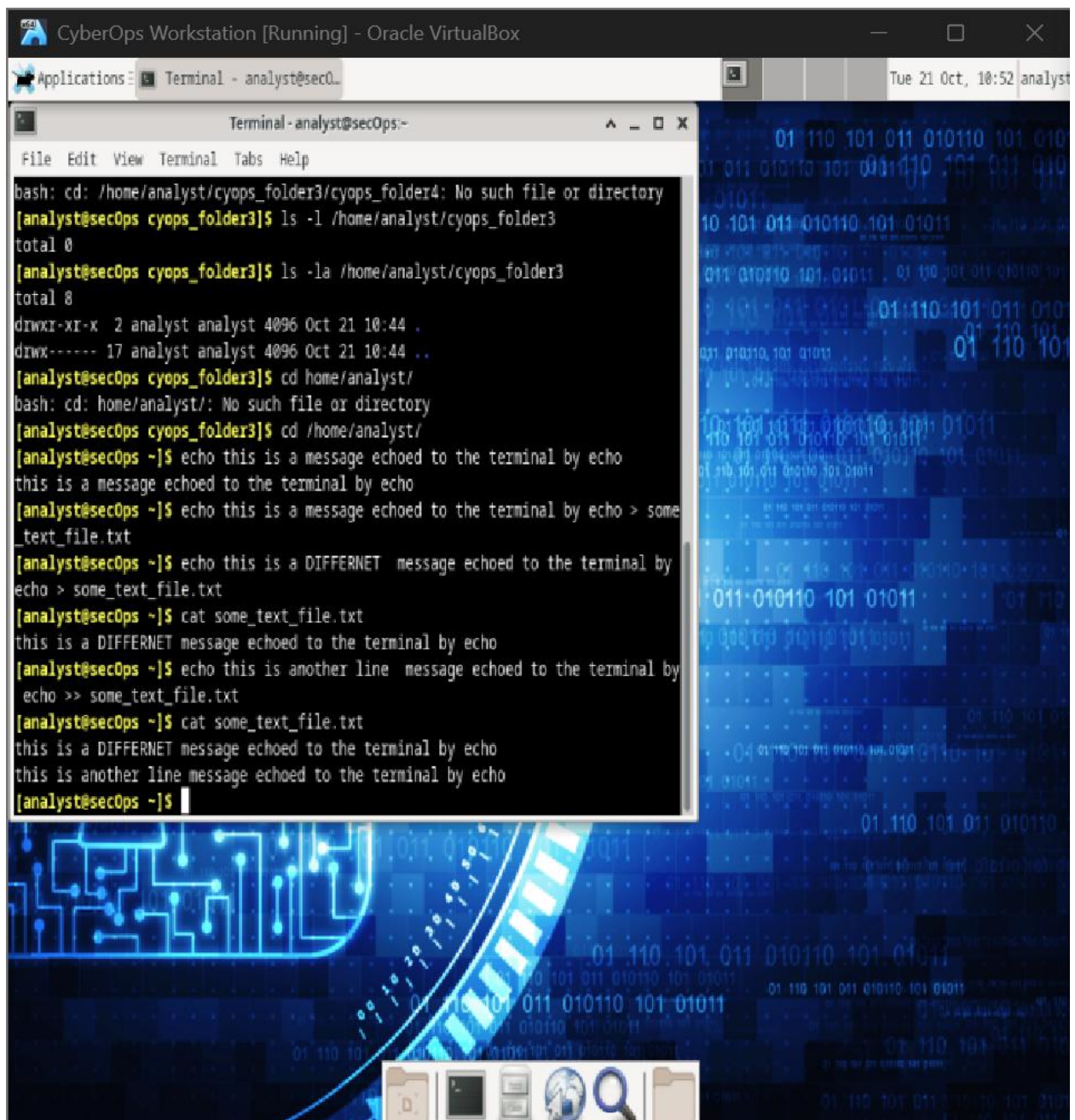


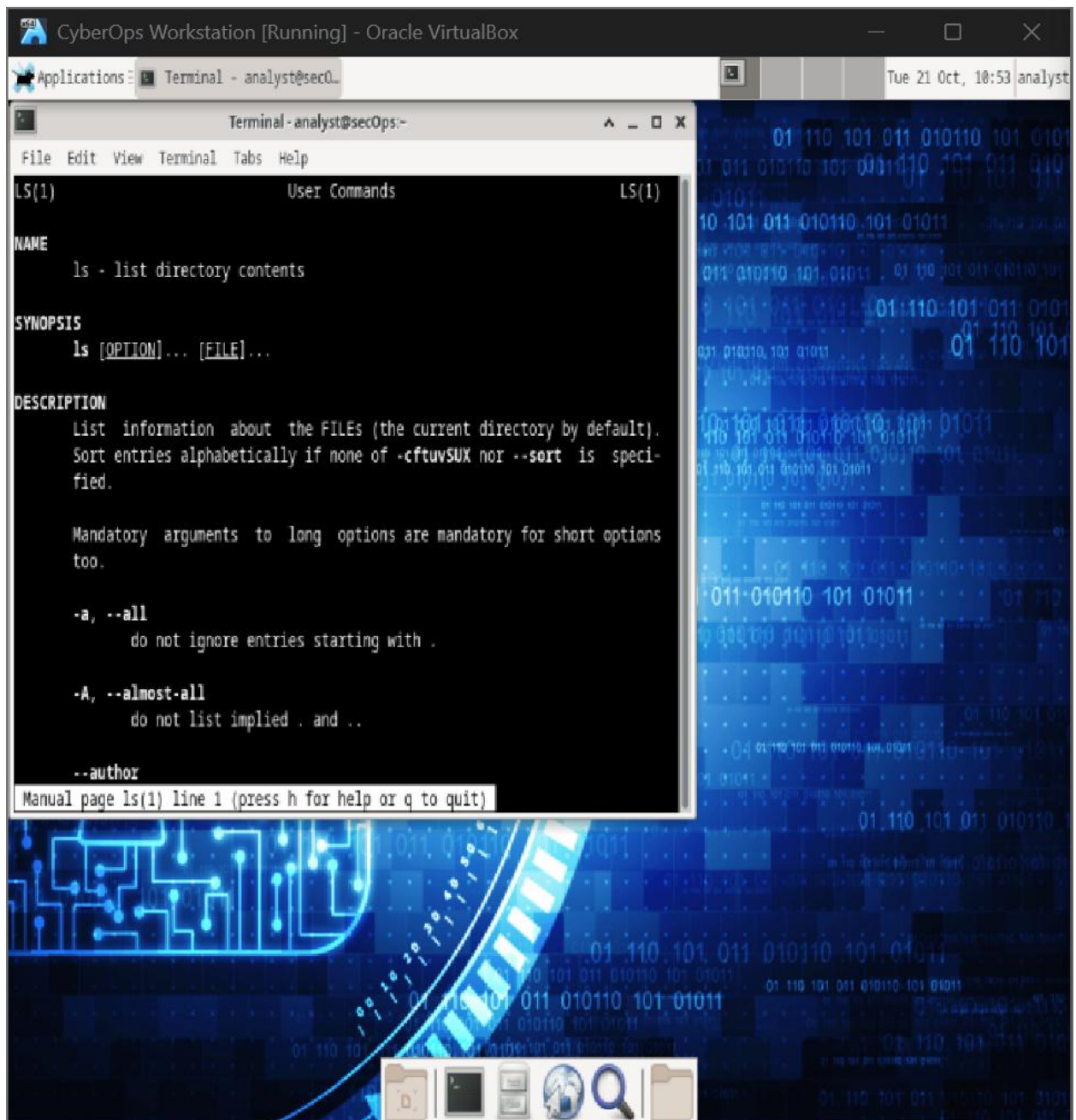


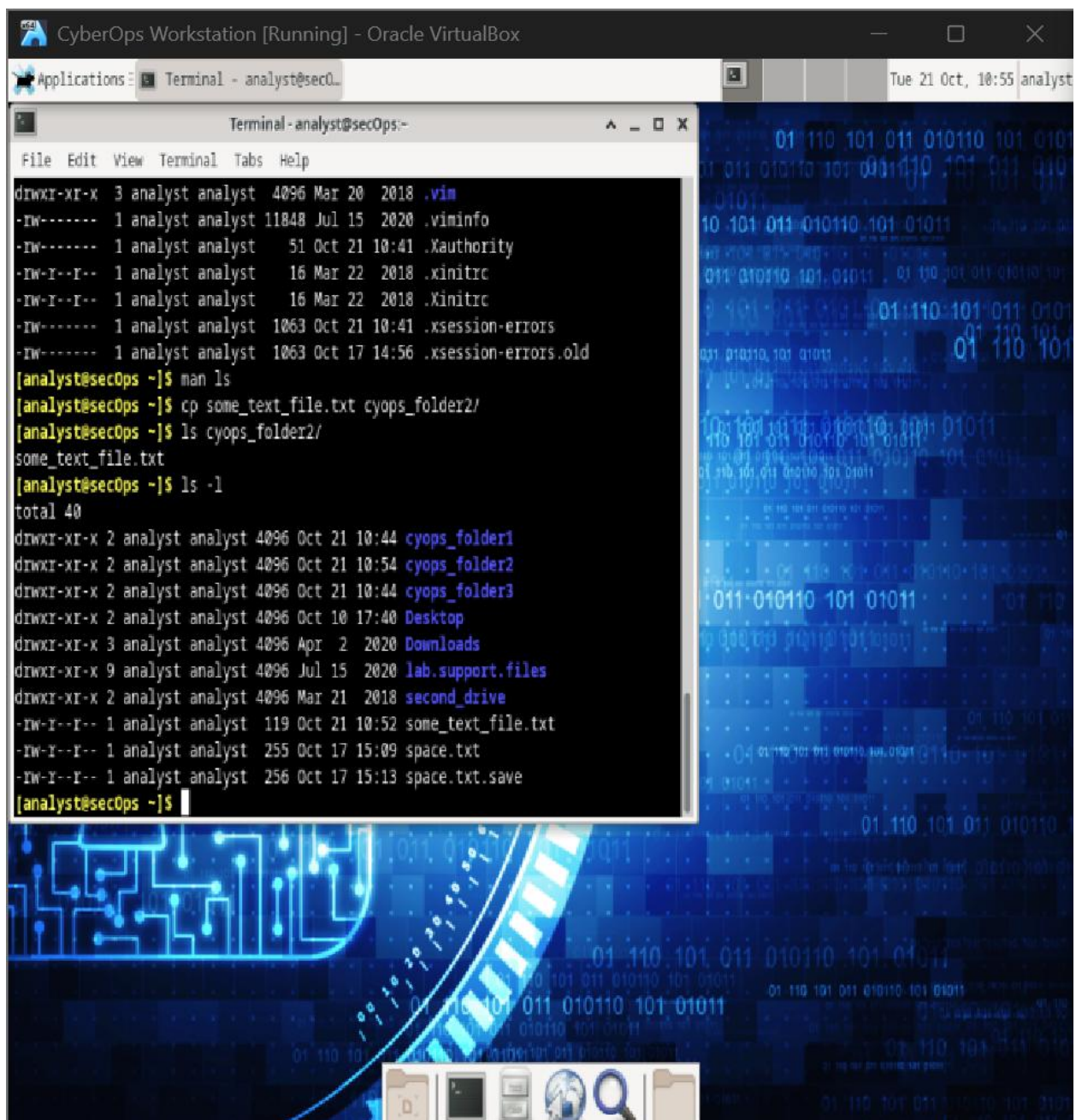


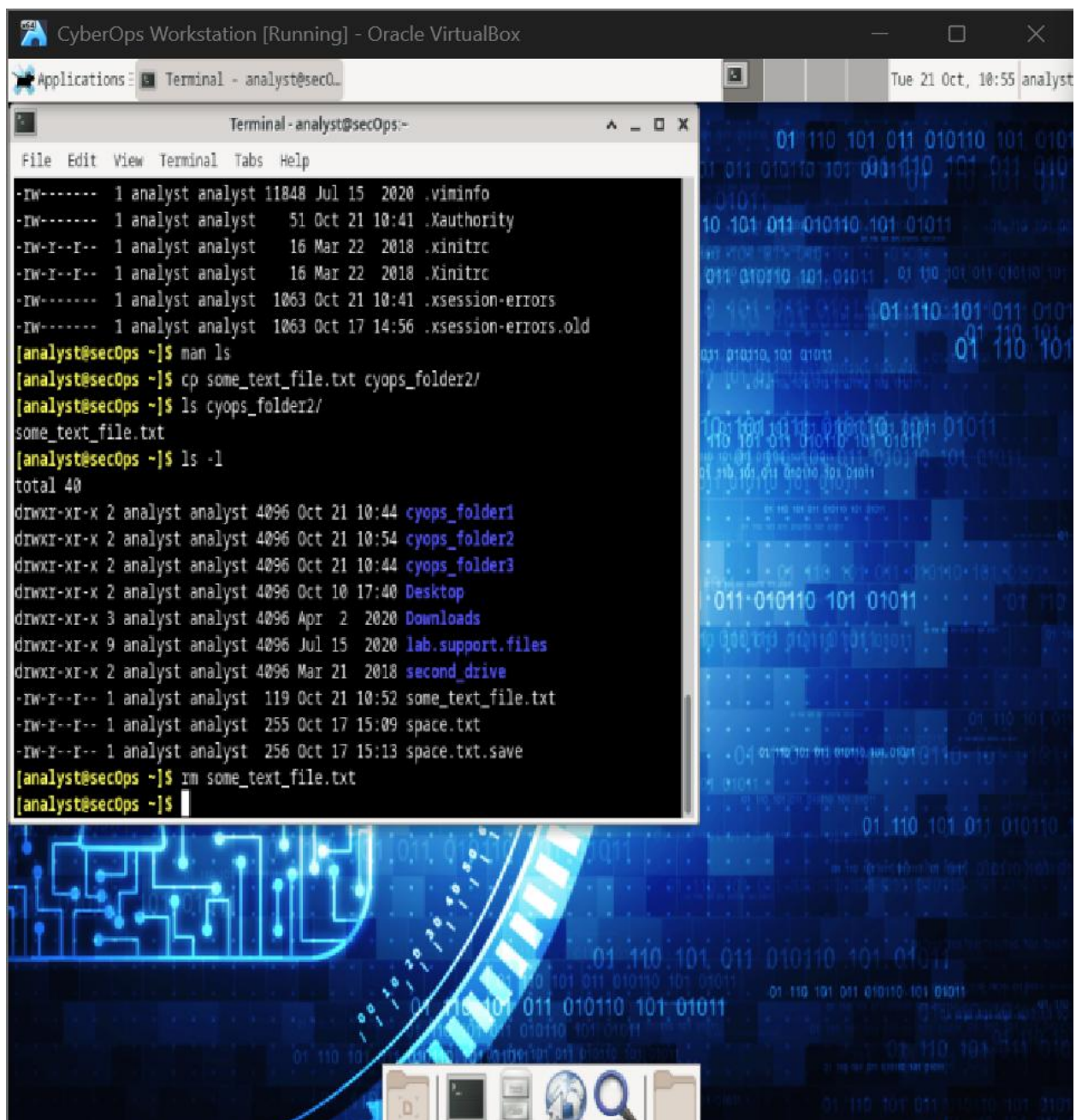













```
CyberOps Workstation [Running] - Oracle VirtualBox
Applications: Terminal - analyst@sec0...
Tue 21 Oct, 10:57 analyst

Terminal - analyst@secOps:~
File Edit View Terminal Tabs Help

drwxr-xr-x 9 analyst analyst 4096 Jul 15 2020 lab.support.files
drwxr-xr-x 2 analyst analyst 4096 Mar 21 2018 second_drive
-rw-r--r-- 1 analyst analyst 119 Oct 21 10:52 some_text_file.txt
-rw-r--r-- 1 analyst analyst 255 Oct 17 15:09 space.txt
-rw-r--r-- 1 analyst analyst 256 Oct 17 15:13 space.txt.save
[analyst@secOps ~]$ rm some_text_file.txt
[analyst@secOps ~]$ ls -l
total 36
drwxr-xr-x 2 analyst analyst 4096 Oct 21 10:44 cyops_folder1
drwxr-xr-x 2 analyst analyst 4096 Oct 21 10:54 cyops_folder2
drwxr-xr-x 2 analyst analyst 4096 Oct 21 10:44 cyops_folder3
drwxr-xr-x 2 analyst analyst 4096 Oct 10 17:40 Desktop
drwxr-xr-x 3 analyst analyst 4096 Apr 2 2020 Downloads
drwxr-xr-x 9 analyst analyst 4096 Jul 15 2020 lab.support.files
drwxr-xr-x 2 analyst analyst 4096 Mar 21 2018 second_drive
-rw-r--r-- 1 analyst analyst 255 Oct 17 15:09 space.txt
-rw-r--r-- 1 analyst analyst 256 Oct 17 15:13 space.txt.save
[analyst@secOps ~]$ mv cyops_folder2/some_text_file.txt
mv: missing destination file operand after 'cyops_folder2/some_text_file.txt'
Try 'mv --help' for more information.
[analyst@secOps ~]$ mv cyops_folder2/some_text_file.txt .
[analyst@secOps ~]$ ls -l cyops_folder2/
total 0
[analyst@secOps ~]$
```

Conclusion

This laboratory exercise successfully demonstrated fundamental Linux shell operations including directory navigation, file manipulation, output redirection, and hidden file management. The hands-on experience reinforced the power and flexibility of the command line interface, essential skills for cybersecurity operations and system administration. Understanding these basic commands provides a foundation for more advanced Linux operations and security analysis tasks.

Commands Summary

Navigation Commands

- `pwd` - Print working directory
- `cd [directory]` - Change directory
- `cd ~` - Return to home directory
- `cd ..` - Move to parent directory
- `cd .` - Stay in current directory

File and Directory Management

- `ls -l` - List files with details
- `ls -la` - List all files including hidden
- `mkdir [name]` - Create new directory
- `cp [source] [destination]` - Copy files
- `mv [source] [destination]` - Move/rename files
- `rm [file]` - Remove file
- `rm -r [directory]` - Remove directory recursively

File Content Operations

- `cat [file]` - Display file contents
- `echo [text]` - Display text or write to file
- `>` - Redirect output (overwrite)
- `>>` - Redirect output (append)

Help and Documentation

- `man [command]` - Display manual page
-